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Explainable AI-Based Personalized Lifestyle Recommendation System for Polycystic Ovary Syndrome (PCOS) Management

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Abstract

Polycystic Ovary Syndrome (PCOS) is a common condition that touches hormonal, metabolic and reproductive health, and often day-to-day well-being as well. Lifestyle management, meaning healthy eating, regular physical activity, enough sleep and stress control, is a core part of long-term care. Yet generic advice rarely fits the habits and needs of each person. Recent work has shown that Artificial Intelligence (AI) and Machine Learning (ML) can help diagnose and predict PCOS, but far less attention has gone to personalized lifestyle support, and many AI models still explain their outputs poorly. This paper reviews existing research on PCOS, AI in healthcare, lifestyle interventions, personalized recommendation systems and Explainable Artificial Intelligence (XAI). Drawing on that literature, it proposes a framework for an Explainable AI-Based Personalized Lifestyle Recommendation System for PCOS Management. The system would take a user's health and lifestyle information and produce tailored guidance on diet, exercise, sleep, stress management and daily routines, along with the reasons behind each suggestion. It is positioned as a decision-support and educational tool, not a substitute for professional medical advice.

Keywords: Polycystic Ovary Syndrome (PCOS), Artificial Intelligence, Machine Learning, Explainable AI, Personalized Healthcare, Lifestyle Recommendation, Digital Health, Decision Support

I. Introduction

A. Background

Polycystic Ovary Syndrome (PCOS) is among the most common hormonal and endocrine conditions in women of reproductive age. It looks different from one person to the next. Some experience irregular menstrual cycles, hormonal imbalance, insulin resistance, weight-related concerns, acne, excess hair growth or infertility, and the condition is also linked to longer-term metabolic and cardiovascular risks. Since symptoms and severity vary so much, managing PCOS usually means paying attention to each person's own health and lifestyle circumstances.

Lifestyle change is a central part of that management. Adjusting eating habits, staying active, sleeping well, managing stress and addressing weight-related behaviours can all improve general health and ease symptoms. But what works well for one person may not suit another, which points to a need for approaches

that organize a person's health information and turn it into more personalized guidance.

AI and ML are being used more and more across healthcare, including for PCOS diagnosis and prediction. Earlier studies have tried algorithms such as Random Forest, Decision Tree, Support Vector Machine and Logistic Regression, and the diagnostic results have been encouraging. Still, the literature leans much more toward identifying PCOS than toward supporting people in managing it over the long term.

Explainable Artificial Intelligence (XAI) offers a way to make AI outputs easier to understand. In a lifestyle recommendation system, that means a user can see which parts of their own profile led to a given recommendation. This need is what motivates the present research on an Explainable AI-Based Personalized Lifestyle Recommendation System for PCOS Management.

B. Problem Statement

Most AI research on PCOS so far has centred on diagnosis and prediction. Lifestyle management, however, depends on many things at once: diet, exercise, sleep, stress and the individual's own health characteristics. Digital health applications often give general advice that overlooks differences between users, and when AI is involved, weak explainability can leave its recommendations hard to follow. The problem this research addresses is how to build a system that pairs personalized lifestyle guidance with clear explanations, while remaining a supportive tool and not a medical diagnostic system.

C. Research Questions

- Can Machine Learning generate effective, personalized lifestyle recommendations for women with PCOS?
- Does adding Explainable AI improve users' understanding of, and trust in, the recommendations they receive?
- Which combinations of health and lifestyle factors have the greatest influence on the recommendations produced?
- How do personalized, explainable recommendations compare with generic lifestyle advice in terms of perceived usefulness?

II. Review of Literature

Author (Year)	Focus / Method	Key Contribution	Gap Identified
Barrera et al. (2023)	Systematic review of ML/AI in PCOS diagnosis	Surveyed algorithms used for PCOS classification	Small datasets; inconsistent diagnostic criteria
Denny et al. (2019)	ML-based detection system (i-HOPE)	Early ML pipeline for PCOS prediction	Diagnosis-only; no lifestyle guidance
Tiwari et al. (2022)	SPOSDS diagnostic system	Smart diagnostic ML system for PCOS	No personalization or explainability
Danaei Mehr & Polat (2022)	Feature-selection techniques with ML	Compared feature-selection methods for PCOS diagnosis	Focused on accuracy, not user guidance
Rahman et al. (2024)	Web-based ML PCOS prediction	Deployed ML models (LR, DT, RF) in a web app	Prediction only; no ongoing lifestyle support

Author (Year)	Focus / Method	Key Contribution	Gap Identified
Umaa Maheswari & Uma Maheswari (2024)	SmartScan PCOS (ML + XAI)	Combined ML with explainability for PCOS prediction	Explainability applied to diagnosis, not lifestyle recommendations
Ghaderzadeh et al. (2025)	Systematic review of AI in PCOS	Reviewed 80 studies on diagnostic/ predictive AI	Limited XAI adoption; calls for research beyond diagnosis
Shahmoradi et al. (2024)	Review of nutritional recommendation systems in PCOS	Found recommendation research far less developed than diagnostic research	Very few personalized, ongoing lifestyle systems
Gautam et al. (2025)	Systematic review of lifestyle interventions in PCOS	Confirmed diet, exercise, sleep, and stress management improve outcomes	Clinical evidence not yet operationalized into AI systems
Sinha et al. (2024)	Review of AI/ML for personalized nutrition	Surveyed AI techniques for tailoring dietary guidance	General nutrition focus, not PCOS-specific
Wang et al. (2025)	Systematic review of AI dietary recommendations	Evaluated AI-generated dietary interventions across conditions	Focused on diabetes/ IBS; PCOS not addressed
Molnar (2022)	Interpretable Machine Learning (textbook)	Foundational reference for interpretable ML methods	Not health-specific; needs domain application

A. Summary of Literature Findings

- AI and ML have been studied extensively for PCOS diagnosis and prediction.
- Lifestyle management is a key part of long-term PCOS care.
- Research on personalized nutrition and lifestyle recommendation is less mature than research on diagnosis.
- Small datasets and inconsistent diagnostic criteria limit how well models generalize.
- Explainable AI can make AI-generated outputs more transparent.
- There is room to bring clinical lifestyle evidence, ML, personalization and XAI together in a single system.

III. Research Gap and Relevance of the Study

A. Research Gaps

The literature review shows a clear imbalance. AI-based PCOS diagnosis is well developed, while AI-supported lifestyle management is not. Most existing studies set out to detect or predict PCOS, and far fewer look at what users can do after diagnosis to manage their lifestyle over time.

The second gap is personalization. Clinical literature supports lifestyle modification, yet digital applications often offer broad recommendations that ignore the user's individual health profile. The third gap is explainability: even when AI informs healthcare decisions, users frequently get no clear account of the factors behind the output.

Dataset limitations come up repeatedly as well. Small samples, mixed populations and differing diagnostic criteria all weaken the generalizability of ML models, and these issues have to be kept in mind when designing and evaluating an AI-based PCOS recommendation system.

B. Relevance of the Proposed Study

The proposed study matters because it moves the focus from simply identifying PCOS to supporting the person living with it. By looking at several aspects of an individual's lifestyle and health profile, the system can organize its recommendations around the areas that count most in daily management.

Including XAI matters too, because recommendations should never seem arbitrary. When each one comes with an understandable explanation, the interaction becomes more transparent and users can see how their information is being used.

C. Expected Contribution

- A framework that combines Machine Learning with personalized lifestyle recommendations for PCOS.
- Integration of Explainable AI so that the reasons behind each recommendation are visible.
- Evidence-based lifestyle guidance as the foundation for generating recommendations.
- A user-oriented web prototype that delivers the personalized recommendations.
- A decision-support approach that complements professional healthcare and does not replace it.

IV. Proposed Methodology

The research follows a straightforward, step-by-step approach. Suitable PCOS and lifestyle data will first be collected and prepared. The health and lifestyle variables will then be examined to identify which factors can support personalization.

Publicly available PCOS datasets from earlier ML studies are candidates, particularly those that include BMI, cycle regularity, physical activity and other relevant health factors. Where information on sleep, stress or dietary habits is missing, supplementary lifestyle survey data can fill the gap and support the personalization component.

Once the data has been preprocessed, ML models including Logistic Regression, Decision Tree, Random Forest and Support Vector Machine will be built and compared. The chosen model will feed into a recommendation layer that draws on the user's characteristics and on evidence-based sources to give lifestyle guidance covering diet, physical activity, sleep, stress management and daily routines.

Explainable AI will then be added, using a technique such as SHAP (Lundberg & Lee, 2017), to show the main factors behind a recommendation. Finally, the approach will be implemented as a web-based interface and evaluated with suitable performance measures. The final dataset, model and XAI settings will be settled during implementation.

A. Proposed System Flow

1. User health and lifestyle information
2. Data preprocessing
3. Feature selection
4. Machine Learning model
5. Personalized recommendation engine
6. Explainable AI layer
7. Personalized lifestyle recommendations
8. User-friendly web interface

B. Recommendation Areas

- Diet and nutrition guidance
- Physical activity and exercise
- Sleep and daily routine
- Stress management
- General lifestyle habits

C. Explainability

The explainability component will show users the main factors behind a recommendation. A suggestion about physical activity, for instance, may be linked to the user's reported activity level, BMI or other relevant inputs. The aim is not to give a medical diagnosis but to make the recommendation process easier to follow.

D. Data Privacy and Ethical Considerations

The system handles personal health and lifestyle information, so data privacy and ethics are treated as core design requirements from the start, not as an afterthought. The approach includes informed user consent before any data is collected, anonymization or pseudonymization of stored health data, use of data only for the stated recommendation purpose, and secure storage suited to sensitive personal information. Consistent with the study's decision-support framing, the system will also display a clear disclaimer at the point of use stating that its output is not a medical diagnosis and does not replace consultation with a qualified healthcare professional.

V. Expected Outcomes

The expected outcome is a working prototype that takes in relevant user information and returns personalized lifestyle recommendations. It should show how Machine Learning can be combined with rule-based or evidence-based recommendation logic and Explainable AI.

Evaluation will take place at two levels. At the model level, standard classification metrics (accuracy, precision, recall and F1-score) will measure the performance of the underlying ML component. At the system level, the consistency of the recommendations and the clarity of the explanations will be assessed, along with user feedback from a small-scale usability evaluation. The final system will be treated as a decision-support and educational application, not a clinical diagnostic tool.

VI. Conclusion

Polycystic Ovary Syndrome affects hormonal, metabolic, reproductive and overall well-being, which makes long-term management important. Healthy eating, regular exercise, good sleep and stress management all help support general health. Because lifestyle needs differ from person to person, though, general advice will not always be suitable.

AI and ML offer a chance to make healthcare support more personal. Existing AI research on PCOS has concentrated mainly on diagnosis and prediction, so there is room for systems that also help users manage their lifestyle over time.

The proposed Explainable AI-Based Personalized Lifestyle Recommendation System aims to fill that gap. It uses an individual's health and lifestyle information to recommend changes in diet, exercise, sleep, stress management and daily routines, and it uses Explainable AI to help users understand why each recommendation is made.

Overall, the study brings Machine Learning, personalized recommendations, clinical lifestyle evidence and Explainable AI together in one healthcare support framework. The system is meant to give users understandable lifestyle guidance and should not be seen as a replacement for professional medical diagnosis or treatment.

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